

CANipulator

Dual CAN-Bus Interface

Documentation



Introduction

The CANipulator is a versatile CAN bus interface designed for bridging, translating, baud-rate matching, and manipulating traffic between two CAN networks. Integrated CAN-FD transceivers support data rates of up to 5 Mbps, while jumper-selectable termination resistors and Molex DuraClik connectors simplify installation and integration.

By connecting two independent CAN networks or devices, the CANipulator enables a wide range of applications, including linking isolated networks, translating messages between peripherals and vehicle networks, integrating devices with otherwise incompatible systems, and performing baud-rate conversion for legacy hardware. Automotive-grade power regulation with reverse-polarity and transient protection ensures reliable operation in demanding electrical environments.

At the heart of the CANipulator is an Espressif ESP32-CS-WROOM-1U module featuring a 32-bit RISC-V processor running at up to 240 MHz. The module provides 384 KB of high-performance SRAM, 16 KB of low-power SRAM, 320 KB of ROM, 8 MB of Flash memory, and 8 MB of PSRAM. Built-in wireless connectivity includes dual-band Wi-Fi 6 (2.4 GHz and 5 GHz), Bluetooth LE 5, Zigbee, and Thread (IEEE 802.15.4).

An optional microSD card slot enables onboard data logging for long-term monitoring, diagnostics, and offline analysis.

Electrical Characteristics

Input Voltage: 5V to 36V (40V absolute max)

Typical Input Voltage: 10V to 15V

I/O Pin Voltage: 3.3V max

Operating current: ~100mA (standard mode)

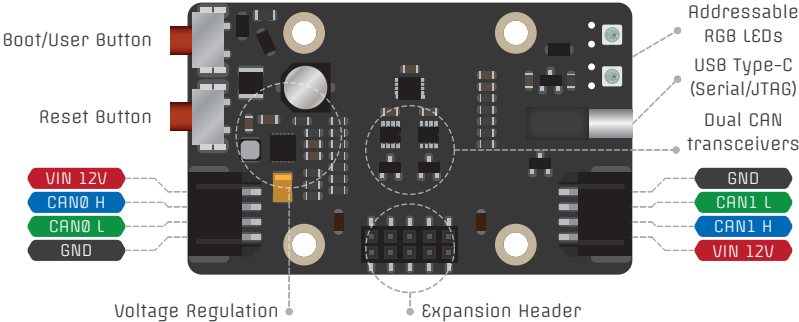
Operating temperature: -40°C to +85°C

Maximum CAN transceiver speed: 5Mbps

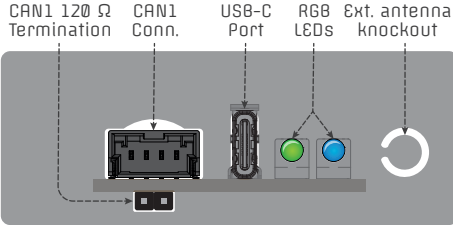
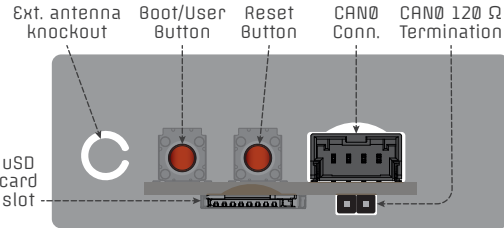
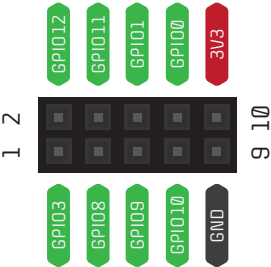
Notes:

- Left/right VIN rails are connected
- CAN0/CAN1 Connector part numbers:
 - Molex DuraClik pigtail 218323-1041; or
 - Molex DuraClik female housing 560123-0400
 - Molex DuraClik housing retainer 560125-0400
 - Molex DuraClik crimp terminal 560124-0101
- Enclosure: Hammond Mfg. 1593K0K

Overview



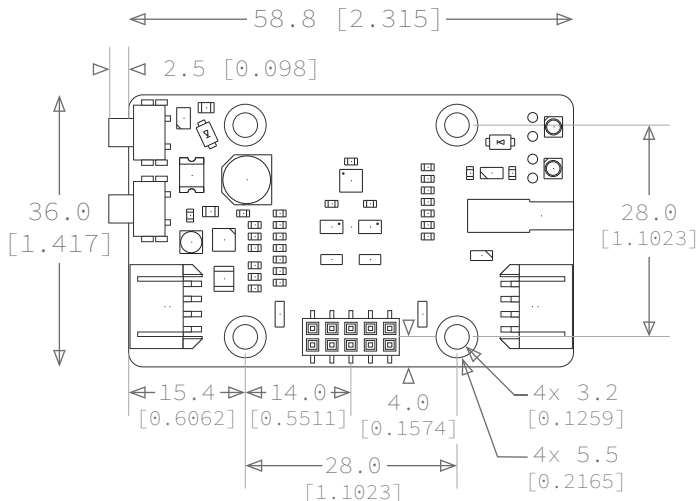
Expansion Header Pinout



I/O Usage & Functions

GPI00	I2C	I2C SDA - Connected to I/O Expander, with 4.7K pull-up
GPI01		I2C SCL - Connected to I/O Expander, with 4.7K pull-up
GPI02		Addressable RGB LEDs data pin
GPI03	CAN-BUS	Unused - Connected to expansion header pin 1
GPI04		CAN0 RX
GPI05		CAN0 TX
GPI06		CAN1 RX
GPI07		CAN1 TX
GPI08		Unused - Connected to expansion header pin 3
GPI09		Unused - Connected to expansion header pin 5
GPI010		Unused - Connected to expansion header pin 7
GPI011		Unused - Connected to expansion header pin 4
GPI012		Unused - Connected to expansion header pin 2
GPI013	USB	USB D-
GPI014		USB D+
GPI023	microSD	uSD MOSI
GPI024		uSD SCLK
GPI025		uSD MISO
GPI026		uSD Insertion Detect pin (low when inserted)
GPI027		uSD CS
GPI028	CAN-BUS	Boot / User button - Can be assigned function post-boot
EXP-I00		CAN0 SIL - Drive high for silent mode, internal pull-down
EXP-I01		CAN0 SHDN - Drive high for shutdown, internal pull-down
EXP-I02		CAN1 SIL - Drive high for silent mode, internal pull-down
EXP-I03		CAN1 SHDN - Drive high for shutdown, internal pull-down

Mechanical



Dimension units: mm [inches], Scale 1:1

Mounting hole pattern designed for: Hammond Mfg. enclosure 1593K8K

Recommended mounting screw size: M3×4mm or #4×1/4" self-tapping (1593ATSS0)

Getting started:

